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10 CFR 50.73

May 1, 2025
Serial: RA-25-0115

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Brunswick Steam Electric Plant, Unit No. 2
Renewed Facility Operating License No. DPR-62
Docket No. 50-324
Licensee Event Report 2-2025-001

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, LLC, is submitting the enclosed Licensee Event Report (LER). This report fulfills the requirement for a written report within sixty (60) days of a reportable occurrence.

This document contains no regulatory commitments.

Please refer any questions regarding this submittal to Mr. Mark DeWire, Manager – Nuclear Support Services, at (910) 832-6641.

Sincerely,

John A. Krakuszeski

John A. Krakuszeski

Enclosure: Licensee Event Report

cc (with enclosure):

Mr. Julio Lara, Acting Regional Administrator, Region II
Mr. Blake Purnell, NRC Project Manager
Mr. Gale Smith, NRC Senior Resident Inspector

NRC FORM 366 U.S. NUCLEAR REGULATORY COMMISSION
(04-02-2024)

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Brunswick Steam Electric Plant (BSEP), Unit 2	[X] 050 [ ] 052 00324	1 OF 3

4. Title
Automatic Actuation of Containment Isolation Valves during Testing

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
03	28	2025	2025	- 001 -	00	05	01	2025		[ ] 050
									Facility Name	[ ] 052

9. Operating Mode	10. Power Level
2	001

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	[ ] 20.2203(a)(2)(vi)	10 CFR Part 50	[ ] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)	[ ] 73.1200(a)
[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.36(c)(1)(i)(A)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)	[ ] 73.1200(b)
[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)	[ ] 73.1200(c)
[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.36(c)(2)	[X] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)	[ ] 73.1200(d)
[ ] 20.2203(a)(2)(i)	10 CFR Part 21	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(A)	10 CFR Part 73	[ ] 73.1200(e)
[ ] 20.2203(a)(2)(ii)	[ ] 21.2(c)	[ ] 50.69(g)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.77(a)(1)	[ ] 73.1200(f)
[ ] 20.2203(a)(2)(iii)		[ ] 50.73(a)(2)(i)(A)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.77(a)(2)(i)	[ ] 73.1200(g)
[ ] 20.2203(a)(2)(iv)		[ ] 50.73(a)(2)(i)(B)	[ ] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(2)(ii)	[ ] 73.1200(h)
[ ] 20.2203(a)(2)(v)		[ ] 50.73(a)(2)(i)(C)	[ ] 50.73(a)(2)(vii)		

[ ] **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Mark DeWire, Manager – Nuclear Support Services	(910) 832-6641

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
[X] No	[ ] Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At approximately 14:09 Eastern Daylight Time (EDT) on March 28, 2025, with Unit 2 in Mode 2 at approximately 1% rated thermal power during reactor startup, an actuation of Group 1 Primary Containment Isolation Valves (PCIVs) occurred during performance of the Unit 2 Turbine Control/Stop Valves Tightness Test procedure. The Group 1 PCIV actuation resulted when the Turbine Stop Valves were opened (with Control Valves remaining closed) while main condenser vacuum was below 10 inches Hg. The PCIVs automatically closed as designed when the Group 1 actuation signal was received.

This event resulted from legacy procedural inadequacies. Corrective actions include procedural revisions to ensure the Turbine Control/Stop Valves Tightness Test procedure is not executed with main condenser vacuum below 10 inches Hg.

There was no impact on the health and safety of the public or plant personnel. This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) due to valid actuation of the Primary Containment Isolation System.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP), Unit 2	■ 050 ☐ 052	00324	2025	- 001	- 00

NARRATIVE

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

Background

Initial Conditions

At the time of the event, Unit 2 was in Mode 2 (i.e., Startup), at approximately 1% rated thermal power.

Reportability Criteria

This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) because it involved actuation of a system listed in 10 CFR 50.73(a)(2)(iv)(B). Specifically, the Unit 2 Primary Containment Isolation System (PCIS) [JM] actuated during this event.

The NRC was notified of this event per 10 CFR 50.72(b)(3)(iv)(A) via Event Notification 57636 at 19:07 Eastern Daylight Time (EDT) on March 28, 2025.

Event Description

At approximately 14:09 Eastern Daylight Time (EDT) on March 28, 2025, with Unit 2 in Mode 2 at approximately 1% rated thermal power during reactor startup following a planned refueling outage, an actuation of Group 1 Primary Containment Isolation Valves (PCIVs) (i.e., main steam line, main steam line drain, and reactor water sample line isolation valves) occurred due to receipt of the Condenser Vacuum Low - Group 1 isolation signal. The Group 1 PCIV actuation resulted when the Turbine Stop Valves were opened in accordance with the Unit 2 Turbine Control/Stop Valves Tightness Test procedure which was in-progress at the time. Main condenser vacuum was less than the 10 inches Hg setpoint when the actuation occurred.

Event Cause

The direct cause of this event was opening Turbine Stop Valves with main condenser vacuum below 10 inches Hg. Per procedure the Condenser Vacuum Low - Group 1 isolation signal is bypassed below 500 psig reactor vessel pressure. Though the Turbine Control Valves remained closed, the Condenser Vacuum Low - Group 1 isolation signal was automatically un-bypassed per design when the Turbine Stop Valves were opened. With main condenser vacuum below the 10 inches Hg setpoint, a Group 1 isolation signal occurred, per design.

This event resulted from legacy procedural inadequacies. In 2019 the Turbine Control System (TCS) [JJ] digital upgrade modification changed the Unit 2 Turbine Control/Stop Valves Tightness Test procedure from a manually controlled test opening just one Turbine Stop Valve poppet, to a TCS automatically controlled test opening all four Turbine Stop Valves. This procedure change did not add a prerequisite to ensure condenser vacuum was greater than the Condenser Vacuum Low - Group 1 isolation setpoint prior to beginning the test.

NARRATIVE

Safety Assessment

There was no adverse impact on the health and safety of the public or plant personnel. The safety significance of this event is minimal. The automatic system actuations occurred as designed and all Main Steam Isolation Valves were reopened within approximately 20 minutes.

Corrective Actions

The Unit 1 and Unit 2 Turbine Control/Stop Valves Tightness Test procedures were revised to include a new prerequisite to ensure condenser vacuum is greater than 10 inches Hg prior to performing the test.

Previous Similar Events

No previous similar events have occurred within the past 5 years in which inadequate procedural guidance resulted in automatic PCIS actuation.

Commitments

No regulatory commitments are contained in this report.